

ENEL700 Tutorial 5

1. Determine the Nyquist Rate and Nyquist Sampling Interval for the following signals: (1) $5 \cos(200\pi t)$ (2) $20 - \sin^2(10^4 \pi t)$ (3) $10 \text{rect}(2 \times 10^3 t)$
2. A sinusoidal voltage signal $v(t) = 20 \sin(2\pi \times 10^4 t)$ volts is sampled using an impulse train $\delta_{T_s}(t)$ of period $T_s = 40 \mu\text{s}$. Sketch the waveform of corresponding double-sided amplitude spectrum of the following signals: (1) $v(t)$ (2) $\delta_{T_s}(t)$ (3) Sampled signal $v_\delta(t)$ (should contain at least three sidebands).
3. A signal that ranges from 0 to +3V must be converted to digital form with a quantisation error of less than 1mV. How many bits per sample would an ADC have to provide to meet this goal?
4. The output of a midrise uniform quantiser of input range $\pm 5\text{V}$ is coded using 8 bits/sample. Determine: (1) the maximum quantisation error; (2) the quantisation error associated with an input sample of value -2.3V; (3) the quantisation noise power; (4) the dynamic range of the system; (5) the SQNR when the input is a speech signal that fully loads the quantiser; (6) the SQNR during a weak passage when the peak value of the input speech signal is only 0.5V.